

# Exploring Cloud Computing Transformations

Report on real-world cloud implementations & solutions

Zane Grose - MDS3000428

ICC104

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# Introduction

As more people utilize the internet to access platforms and applications, companies are increasingly needing to provide their systems with sufficient computing power, storage, security, availability, accessibility, and recovery options. As companies migrate past on-premises hardware and software, it is important for them to determine what type of cloud computing service and deployment models fit their business goals.

The cloud allows companies to enhance all these features of a business' system's needs, utilizing three core service models (IBM , n.d):

- **Infrastructure as a Service (IaaS):** Provides on-demand access to cloud-hosted physical and virtual servers, storage, and networking. Allowing full configuration over the infrastructure.
- **Platform as a Service (PaaS):** Provides on-demand access to a complete, ready-to-use, cloud-based platform for developing, running, maintaining, and managing applications.
- **Software as a Service (SaaS):** Provides on-demand access to ready-to-use, cloud-hosted application software.

Deployment models accompany these service models, and consist of four main models: public, private, hybrid, and multi clouds. These models determine who can access, manage, and scale specific infrastructure, software, and resources.

These cloud deployment and service models are imperative for businesses of all sizes to adapt if they want to grow efficiently. The cloud provides companies with diverse options to scale their business, manage accessibility, allocate computing resources, provide around the clock availability, and implement robust recovery options to fit business needs.

# Netflix

Netflix is a streaming entertainment platform that provides a variety of movies, tv shows, and games. The cloud is critical in the home-entertainment industry to deliver appropriate computing resources, storage, scalability, infrastructure, and content distribution.

Netflix faced overheads via running their own emailing servers (AWS, 2021), along with suffering from a 'major database corruption' in 2008 (Izrailevesk *et al.*, 2016), this corruption event was due to having vertically scaled servers, that enabled a single-point of failure.

Amazon SES as a PaaS was implemented for emailing services, enabling Netflix to efficiently contact customers and employees across 190 different countries, removing the need for Netflix to supply their own hardware, software, and ISP configurations for emailing services.

Netflix also introduced Amazon S3 IaaS (AWS, 2022), which allows them to store huge amounts of data, necessary for storing high-quality video files and moving assets

across systems. This reduces costs by eliminating on-premisis hardware and software, only utilizing necessary storage services.

In 2012, Netflix launched a cloud-based VFX studio. Building on top of NICE DCV (IaaS) and utilizing Amazons EC2 G4 instances (AWS, 2021) allowed employees to be creative from remote locations, on responsive systems, while further reducing costs.

The hybrid cloud infrastructure (AWS, 2022) implemented fit Netflix's business needs well, allowing the utilization of already implemented on-site systems, combining cloud resources into the system allowing for scaling without eliminating already implement infrastructure.

## Tiger Brokers

Tiger Brokers (TB) is an online brokerage firm, supplying investors across the world with multiple investing options across the US, Hong Kong, Singapore, and Australia. The financial technology (fintech) industry by design requires real-time statistics, ultra-low latency, stability, information security, and secure infrastructure, where the cloud provides solutions.

In April 2020, the Financial Markets Authority (NZ) issued a warning to TB (FMA, 2020) regarding verification of customer information and assets. Three weeks later, TB implemented a cloud-hosted solution run on Amazon EC2 servers, the SaaS ARTEMIS created by Cynopsis (Cynopsis, 2020). This provided TB with a software to automatically conduct risk assessment and screening, automatically verify users and track high-risk customers, ensuring that regulatory compliance with the FMA is met.

TB also identified network connectivity and performance issues. They decided to implement Alibaba's cloud solutions, Global Accelerator, Network Address Translation, and Elastic IP Address. These solutions provide real-time service continuity, accessibility, and availability by providing cross-zone disaster recovery, scheduling user traffic, and higher throughput capabilities for network connectivity across multiple geographic locations.

TB incorporated a hybrid-cloud infrastructure, ensuring sensitive system-critical information is secure on private clouds, while utilizing the public cloud to enhance computing, networking, and connectivity options. This decision to adopt a hybrid-cloud deployment model was well placed as it fits the needs for security, stability, and accessibility.

## Department of Internal Affairs

The New Zealand Department of Internal Affairs (DIA) is a multi-agency organization that provides services like RealMe. RealMe is a centralized online platform where residents can opt-in to verify their identity with one login, across one-hundred and sixty-three government services utilizing fifty-six public agencies. Cloud computing is important for government agencies considering the amount of sensitive data they

manage, making it necessary to provide availability, maintenance, storage, and secure information.

The maintenance of the pre-cloud system was extremely costly, requiring time-consuming custom builds, updates to network infrastructure security, and requiring manual password contributing to the millions in costs. The DIA's government agencies were also concerned about the security surrounding a cloud implementation, which required DIA to foster trust to adapt the change towards the cloud (Microsoft, 2021).

RealMe (DIA) partnered with UNIFY solutions, an advanced specialist in information protection, which already had success managing NZ Police and NZ Ministry of Education cloud services with Microsoft Azure. This helped establishing trust between the DIA and government agencies they work with.

Utilizing Entra ID (Identity as a Service) allowed the DIA to easily integrate their old systems with the cloud, enhance security, enable scalability, and further integration with established apps and databases, lowering the total cost of operations (TCO). Entra ID also provides a built-in self-service password reset feature, which removed the need to allocate resources to manually reset passwords.

DIA kept their on-premises solutions to store their users verified identity data but utilize Entra ID to access the RealMe platform. This utilization of the public cloud lowered TCO but could have been accompanied with a private cloud solution. This private cloud could store sensitive information, while keeping it secure.

## Etihad Airways

Etihad being one of two premium airline providers in the UAE want to provide their customers with the best user-experience and low costs. The cloud is an important integration in the travel industry to allow businesses to easily scale their operations, provide seamless user-experiences, generate useful analytics, and reduce operational costs.

Before integrating to the cloud, Etihad used an 'off-the-shelf' check-in application, which was no longer fitting user-requirements. In attempt to solve this, they created a mobile app utilizing API-driven backend technology, which passengers preferred. Even though this solution was a clear option for passengers, the API-driven app was becoming too complex to upkeep with the level of personalization Etihad was trying to achieve (IBM Cloud, n.d).

Etihad decided to use IBM cloud as part of their solution within the API economy layer. This allowed Etihad to develop cost-effective, cloud-native applications and develop within a mature DevOps toolchain. The integration of Common Travel Services (CTS) deployed on IBM cloud Kubernetes also assisted in reaching business goals. CTS provides a cluster of compute hosts (Cluster as a Service) to deploy and manage containerized applications, allowing Etihad to move workloads around the platform and assist in development within mobile channels and kiosks.

The decision to utilize a multi-cloud structure allowed Etihad to quickly develop applications and integrate them within their new cloud-based services. This kept sensitive information safe, allowed generation of useful customer information, and allowed integration with the microservices architecture that was already in place.

## Conclusion

Common factors between these case studies are the need to increase computing power, network resources, storage capacity, reduce costs and integrate with legacy systems. Investigating further into cloud technology such as edge computing and AI-cloud management, and containers (used by Etihad) will enhance the transformation to the cloud. Finally, ensuring that the transformation to the cloud is accompanied by keeping up with privacy, regulation, DevOps, and security standards is crucial for business success and to maintain trust and meet compliance requirements.

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